

PHAN HUU KHA

Aspiring Backend / Systems Engineer

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PROFILE

Software Engineering Student at Sai Gon University with a dedicated focus on Backend Architecture, Distributed Systems, and DevOps. Experienced in designing high-throughput event-driven messaging pipelines, in-memory caching filters, and zero-trust self-hosted homelab infrastructure. Seeking a Backend / DevOps Internship or Fresher role to contribute to resilient, production-grade systems.

EDUCATION

Sai Gon University

Engineer's Degree in Software Engineering

Sept. 2023 – June 2028

Current GPA: 3.4 / 4.0

TECHNICAL SKILLS

Languages: Java (17/21 LTS), SQL, JavaScript (ES6+), Bash Shell

Backend Frameworks: Spring Boot 3.5, Spring Security 6 (JWT / OAuth2 / RBAC), Spring Data JPA/Hibernate, WebSocket (STOMP), SSE, RESTful APIs

Databases & Queues: PostgreSQL, MongoDB, Redis Stack (RedisBloom / Pub/Sub), Apache Kafka (KRaft / Avro)

DevOps & Infrastructure: Docker Compose, Google Jib, Jenkins CI/CD, GitHub Actions, Tailscale (Funnel), Nginx, Antigravity CLI

Testing & Architecture: JUnit 5, Mockito, MockMvc, Swagger / OpenAPI 3.0, Event-Driven Architecture (EDA), Distributed Clustering

PROJECTS

ChatWeb – Real-Time Distributed Messaging Platform

[🔗 Code](#)

Role: Backend Developer & System Designer | Java 21, Spring Boot 3.5, Kafka, RedisBloom, PostgreSQL, MongoDB, STOMP, Nginx, Docker Compose

- Engineered an event-driven Write-Behind pipeline using **Kafka (KRaft, Avro)** with dual consumers: a push consumer (<10ms latency) and a batch consumer grouping up to 200 messages into MongoDB with Dead Letter Topic (DLT) error isolation.
- Scaled WebSocket clustering across multi-instance nodes using a **Redis Hash** session directory and node-targeted Redis Pub/Sub channels, completely eliminating broadcast storms.
- Architected robust auth via **Spring Security 6** (JWT, HttpOnly refresh cookies, OAuth2, RBAC), STOMP handshake validation with Redis token blacklisting, and atomic PostgreSQL `token.version` for instant multi-device session revocation.
- Prevented DB exhaustion via **RedisBloom Cuckoo Filter** (O(1) account preflight), atomic Redis Lua rate limiters, 5s presence debounce queue, and MongoDB `$max` atomic read receipts.
- Orchestrated 10+ containers via **Docker Compose** with healthcheck chains, upstream TLS via private Root CA, automated CI/CD via **GitHub Actions & Google Jib**, and API validation via **Swagger / OpenAPI 3.0**.

Confession-Wall – Homelab Server Infrastructure & Community Platform

[🌐 Live](#) | [🔗 Code](#)

Role: Fullstack Engineer & DevOps Owner (Personal Lab) | Bare-Metal Linux, Spring Boot, PostgreSQL, Jenkins, Tailscale, Docker

- Self-hosted and provisioned a physical **Ubuntu bare-metal server** running containerized production workloads (Spring Boot, PostgreSQL, Nginx, Jenkins) under explicit CPU/RAM resource quotas.
- Established Zero-Trust public ingress via **Tailscale Funnel**, tunneling public HTTPS traffic over encrypted WireGuard directly to Nginx without port forwarding or exposing residential IP.
- Constructed an end-to-end 5-stage declarative **Jenkins CI/CD Pipeline** inside Docker (`/var/run/docker.sock`): automated H2 Maven testing, multi-stage builds, and rolling updates with curl healthchecks.
- Built real-time confession streaming via **Server-Sent Events (SSE)** synchronized with Spring `afterCommit` transaction hooks, guaranteeing zero ghost broadcasts on rollback.
- Achieved sub-millisecond (<1ms) case-insensitive search across confession posts by indexing PostgreSQL with **GIN pg_trgm** trigram extensions, with architecture scaffolding accelerated via **Antigravity CLI**.

Other Projects (Available on Portfolio): *Secure-Chat-Web* (E2EE AES-GCM/RSA, Sparse Merkle Tree, Ethereum) | *Smart-Doc AI* (Django 5 ASGI, FAISS + BM25 Hybrid RAG, Cross-Encoder Re-ranking).

LANGUAGES

Vietnamese: Native | **English:** Professional working proficiency (technical documentation, specifications)